Computer Science and Engineering Department
B.Tech CSE Sem VII
4CS505ME25– Cloud Native Applications and DevOps
Year 2026-2027

Course Code	**Subject Name**	**[3-0-2-4]**
4CS505ME25	Cloud Native Applications and DevOps	**[3-0-2-4]**

List of practical

Sr. No	Title	Hours	CO
1	Learn Git related commands about how to create project and manage it on Git environment locally as well as on GitHub.	2	3
2	Containerizing a Web Application Using Docker.	2	3
3	Implement a Multi-Container Application Using Docker Compose.	2	3
4	Deploy a Cloud-Native Web application on AWS ECS using ECR and Create Lambda function to invoke ECS API.	2	3
5	Building a Microservice-Based Application and Deploying Applications on Kubernetes.	2	4
6	Implement Service Discovery, Load Balancing in Kubernetes.	2	3
7	Implement configuration management in Kubernetes using ConfigMaps and Secrets and deploy a containerized application that consumes both non-sensitive and sensitive configuration data.	2	4
8	Set up monitoring tools like Prometheus and Grafana to monitor the performance and health of a deployed application. Incorporate logging solutions like ELK (Elasticsearch, Logstash, Kibana) to collect and analyse logs.	2	4
9	Implement DevSecOps Pipeline with GitHub Actions and Security Scanning Tools.	2	4
10	Implement a Continuous Integration and Continuous Deployment (CI/CD) pipeline for a cloud-native application using GitHub, GitHub Actions, Docker, Docker Hub, and Kubernetes.	2	3